

DATASETS, BENCHMARKS, AND PROTOCOLS

Deidentifying Medical Documents with Local, Privacy-Preserving Large Language Models: The LLM-Anonymizer

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Abstract

BACKGROUND Medical research with real-world clinical data is challenging as a result of privacy requirements. Patient data should be anonymized before analysis in research studies. Anonymization procedures aim to reduce the reidentification risk below a certain threshold, while maintaining the usefulness of the data for research purposes. However, in the context of medical text, these procedures are notoriously hard to automate and, therefore, are not scalable. Recent advancements in natural language processing (NLP), driven by the development of large language models (LLMs), have markedly improved the automatic processing of unstructured text.

METHODS We hypothesize that LLMs are highly effective tools for extracting patient-related information, which can subsequently be used to remove personal information from medical reports, while at the same time preserving information required for downstream research purposes. To test this, we conducted a benchmark study using eight local LLMs (Llama-3 8B, Llama-3 70B, Llama-2 7B, Llama-2 70B, Llama-2 7B Sauerkraut, Llama-2 70B Sauerkraut, Mistral 7B, and Phi-3 Mini) to extract and remove patient-related information from a dataset of 250 real-world clinical letters.

RESULTS Our results demonstrate that our LLM-Anonymizer, when used with Llama-3 70B, achieved a success rate of 99.24% in removing text characters carrying personal identifying information. It missed only 0.76% of text characters with identified personal information and mistakenly redacted 2.43% of characters.

CONCLUSION We provide our full LLM-based Anonymizer pipeline under an open-source license. Its user-friendly web interface operates on local hardware and requires no programming skills. This tool has the potential to facilitate medical research by enabling the secure and efficient deidentification of clinical free-text data on-site, thereby addressing key challenges in medical data sharing. (Funded by German Federal Ministry of Education and Research, CAMINO, 01EO2101, and the European Research Council, ERC; NADIR, 101114631.)

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Background

The digitization of medical records, which include clinical letters, reports, and various other forms of patient data,¹ has grown substantially in past years.^{2,3} Electronic Health Records (EHRs) are moving toward unified digital repositories that encapsulate each patient's complete health care journey, with the goal of creating integrated platforms accessible to health care professionals on a national scale, and even between international health care systems, for example, within the European Union.⁴ This centralization of data potentially enhances utility for clinical and research purposes and, overall, facilitates more seamless and continuous health care for providers and patients. It also opens new avenues for a text-data-driven approach to medical research⁵ by enabling the systematic extraction of structured information from the growing volume of digital text-based hospital data using natural language processing (NLP) tools.

Recently, the introduction of large language models (LLMs) has markedly expanded the capabilities of NLP in medicine.⁶ While developing conventional NLP models requires specific training on annotated text in order to retrieve information, generalist LLMs can solve similar tasks without further fine-tuning. Contemporaneous LLMs possess some medical knowledge⁷ and demonstrate excellent performance in extracting structured information from unstructured medical texts.⁸⁻¹¹ Simultaneously, the relevance of applying artificial intelligence (AI) algorithms to analyze diverse types of unstructured medical data, such as imaging, text, and genomic data, is increasing.¹² This trend is culminating in the development of general medical foundation models.¹³ All of these new technologies rely on the availability of accessible, real-world, large-scale medical datasets.^{13,14}

The integration and use of such large-scale datasets present multiple challenges, such as ensuring patient privacy. The sensitive nature of patient data precludes the indiscriminate sharing of free text between institutions, and often even within one institution, as a result of the risk that potential breaches could compromise personally identifiable information (PII) during research procedures. Anonymizing patient data before using it for research is, therefore, the gold standard to preserve privacy in research, but automated anonymization of medical documents as clinical letters is highly challenging.^{15,16} The process of anonymization exists on a continuous spectrum; personal information should be removed until the risk of reidentification falls

below a certain threshold, while at the same time preserving the information of interest. Thus, the degree of anonymization depends on the intended downstream use of the data. In this study, we address the problem of removing identifiable information from medical letters (e.g., names, date of birth, patient ID, addresses), while preserving information required for downstream research studies (e.g., diagnosis, surgery date). This approach also aligns with key international regulations and frameworks for clinical trials, such as the European Union — Clinical Trials Regulation (EU-CTR) and the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use — Good Clinical Practice (ICH-GCP).

Medical document formats vary between hospitals and even departments, making it challenging to find a universal software solution for anonymization. Current approaches often rely on time-consuming, expensive, and imprecise manual work¹⁷ or rigid named entity recognition (NER) keyword search, which requires costly software.¹⁸ We hypothesized that LLMs can be used as tools to automate the anonymization of medical documents due to their capabilities of interpreting context and subtle expressions in natural language. Based on this, we developed an open-source anonymization tool based on LLMs that can be run entirely on local consumer hardware. Our tool includes a browser-based front end with an LLM-enabled back end to process text-based medical records, including commonly used data formats such as portable document files (PDFs), text (TXT) files, and Microsoft Word documents containing patient reports. It extracts text data from these files, uses the LLM to identify PII, and then removes it, producing deidentified reports along with a table cataloging the extracted PII. We systematically and quantitatively evaluated the capabilities of several LLMs, including Llama-3, Llama-2-based models, and Mistral models, which can be employed at the point of care. Our solution represents a practical advance in privacy-preserving technologies for health care data management — a step toward using the vast potential of digital health records for research and clinical improvement without compromising patient privacy. Designed for simple accessibility, our tool requires no advanced programming skills from its users and is released as an open-source tool.

Methods

The goal of our study was to investigate the feasibility of using an LLM-based pipeline for zero-shot anonymizing medical documents without the need for dedicated LLM training. We benchmarked this pipeline on real medical

text data and various locally deployable LLMs. In addition, we developed a user-friendly, open-source front end that allows anonymization of medical documents, the evaluation of the deidentification with an annotated dataset, and easy document review.

BENCHMARKING AN LLM-BASED PIPELINE FOR ANONYMIZATION

Data Collection

The data used for the benchmarking experiments included German clinical letters (n=250) from the Department of Medicine I, University Hospital Carl Gustav Carus Dresden. Clinical letters were from 250 randomly selected patients who had been treated between September 2004 and January 2023. A plethora of residents and senior physicians from internal medicine wrote the letters in different combinations. The letters' length ranged from short, two-page reports to long letters with up to 24 pages (Fig. S6 in the Supplementary Appendix), and their focus was on general internal medicine, gastroenterology, hematology, and oncology. They encapsulate free text, tables for laboratory results, endoscopic intervention reports, radiology reports, and consultative reports from other specialties. All research procedures were conducted in accordance with the Declaration of Helsinki. Ethics approval was granted by the ethics committee of Technical University Dresden, reference number BO-EK-400092023.

Data Preprocessing and Model Selection

All 250 clinical letters were collected in PDF format. We extracted the text from the PDF and divided it into smaller text chunks of 4000 characters each. This chunk size was chosen due to the limited context window size for prompt and text chunk processing of the benchmarked LLMs, which allowed processing with all models. For LLM inference, we set up llama.cpp, an open-source C++ library that allows inference of LLMs on local hardware.¹⁹ We tested a small (7 billion parameters [7B])²⁰ and a larger model of Llama-2 (70 billion parameters [70B]),²¹ a Llama-2 model that has previously been fine-tuned on the German language, Llama-2 Sauerkraut 7B and 70B,²² as well as Llama-3 with 8 billion parameter size (8B)²³ and Llama-3 70B.²⁴ We extended the Llama models with Mistral 7B²⁵ and Phi-3 Mini.²⁶ To enable use on comparatively low-resource hardware, we employed only quantized models (four- or five-bit-quantized models), which are smaller than unquantized LLMs. The models were instructed to retrieve PII from each chunk of text programmatically. To ensure a consistent output

structure, we used grammar-based sampling, an approach that allows predefining the output structure of the LLM in a JavaScript Object Notation (JSON)-formatted object via the llama.cpp framework. (Further details can be found in the Supplementary Appendix.)

Definition and Extraction of Patient Identifiers

In this article, we defined PII as follows based on their relevance for clinical research and according to the findings in the clinical letters. PII includes the patient's first name, second name, birth name, date of birth, street, house number, zip code, and patient ID. This definition is not aligned with the protected health information definition by the Privacy Rule of the Health Insurance Portability and Accountability Act of 1996 (HIPAA) by the United States, as this study was meant to demonstrate a proof of concept. Our tool is designed to assist in the deidentification of specific patient-identifying information necessary for clinical research, not for the public disclosure of sensitive data. This is designed to comply with the European General Data Protection Regulation (GDPR), which serves as the European counterpart to HIPAA. The difference between HIPAA anonymization and GDPR anonymization lies primarily in their approaches to defining anonymity — HIPAA anonymization requires the removal of specific data elements, whereas GDPR focuses on ensuring individuals cannot be reidentified. The prompt (translated from German) is displayed in Figure S1B. The same prompt was used for all models.

Evaluation of Model Results

To establish a reference ground truth for our experiments, all 250 clinical letters were manually annotated by two medical doctors, removing all PII. Based on these two annotations a consensus annotation was created, which was used for all evaluations. Annotations were performed on the original PDF documents using the open-source annotation tool Inception and exported in JSON format.²⁷

The LLM's output consisted of structured JSON files containing predictions for all requested PII. These outputs were redacted in the original documents using exact string matching. Although a fuzzy matching option was available, allowing for partial matches or minor character deviations, our evaluation focused on exact matches to ensure the most cautious approach. Subsequently, the human annotations were programmatically compared with the redacted PDF documents on a character-by-character basis, with matches and mismatches counted to determine performance metrics. Performance metrics for our results were calculated

using macro averages, where metrics were computed individually for each clinical letter and, subsequently, averaged across all letters, yielding a comprehensive assessment of overall performance.

LLM-ANONYMIZER PIPELINE ACCESSIBILITY AND OPERATION

We developed a fully automated pipeline, the LLM-Anonymizer, available on GitHub at <https://github.com/KatherLab/LLMAnonymizer-Publication>. In our pipeline, various open-weight LLMs were prompted to extract the personal identifiers (patient name, first name, last name, birth name, birth date, address, and patient ID) from pre-processed clinical letters. The extracted information was redacted within the PDF file using exact string matching. After this report redaction, we calculated performance metrics by comparing the pipeline results with manual annotations by two observers (Table 1). The README file guides users through setting up the pipeline, which can then be run from the terminal and accessed via a browser. It simplifies data anonymization from real-world medical documents through a user-friendly interface with four main tabs (Fig. 1):

1. Preprocessing: Upload raw data from various formats (e.g., PDF, TXT), preprocess it for use in the pipeline, and generate a zip file for storage (see Fig. S1A).
2. LLM information extraction:
 - a. Model configuration — Define LLM model parameters, select hyperparameters, adapt the prompt, and specify items for identification in grammar.
 - b. Run anonymization pipeline — Hit the Run Pipeline button. The model starts, and a progress bar tracks the processing status and estimated time remaining (see Fig. S1B).
3. LLM results: Download the processed files, including original, redacted, and preprocessed documents, which are stored in a zip file (see Fig. S3).
4. Report redaction: Upload the output in a zipped file and the annotated datasets for comprehensive reporting. Obtain global metrics, micro and macro scores, and a confusion matrix. Individual document analysis reveals false-negative rates (FNRs), ensuring a thorough redaction of PII. We included an option for fuzzy matching instead of exact matching, which allows flexibility in removing similar words that may account for spelling variations (Fig. S2A).

Additional details on the pipeline steps can be found in the Supplementary Appendix.

Results

PERSONAL IDENTIFIER REDACTION

All performance metrics are displayed in Table 1. We found high accuracy for all tested models (Llama-3 8B mean accuracy [mA]: $98.2 \pm 1.2\%$; Llama-3 70B mA: $98.2 \pm 1.2\%$; Llama-2 Sauerkraut 7B mA: $98.9 \pm 0.9\%$; Llama-2 Sauerkraut 70B mA: $98.1 \pm 0.7\%$; Mistral 7B mA: $97.9 \pm 1.9\%$). Percentages describe absolute character matches from the LLM-Anonymizer and the consensus of two human experts. The highest sensitivity was achieved by Llama-3 models (Figs. 2A and 2C), with Llama-3 8B performing at the same level as Llama-3 70B, thus indicating a high fraction of PII being correctly redacted (sensitivity 8B: $99.4 \pm 2.5\%$; 70B: $99.2 \pm 4.0\%$). Phi-3 Mini and Llama-2 7B Sauerkraut showed the lowest sensitivities, with $91.7 \pm 10.5\%$ and $84.2 \pm 10.7\%$, respectively. Llama-2 70B Sauerkraut showed a sensitivity of $99.1 \pm 2.3\%$. The specificities of Llama-3 70B and Llama-2 Sauerkraut 70B were 98.1%. Llama-2 Sauerkraut 7B presented a specificity of 99.2%, Llama-2 7B of 97.4%, Llama-3 8B of 98.1%, Llama-2 70B of 97.4, Mistral 7B of 97.9%, and Phi-3 Mini of 96.7%.

The FNR, which is defined as the proportion of personal identifiers where redaction was missed by the LLM-Anonymizer, was lowest for the Llama-3 70B model. This model also had the highest number of clinical letters without any personal identifiers that were missed during the redaction process ($n=230$) (Fig. 2B). The models Llama-3 8B (FNR $0.58\% \pm 2.5\%$) and Llama-2 70B Sauerkraut (FNR $0.95\% \pm 2.9\%$) exhibited the next best performance, with 224 and 217 letters, respectively, without any instances of missed redactions. Overall, the models in the Llama-3 group performed best at redacting sensitive information. This is illustrated in Figure 3, which shows the FNR of all labels per model. Larger models with more model parameters performed better with a FNR distribution toward zero. Llama-3 8B showed the best performance among the smaller parameter-sized models.

The patient's first name and date of birth were consistently the most accurately redacted personal identifiers across all models. All information for the patient's date of birth was deidentified with no missing characters in 245 out of 250 letters for Llama-2 70B, Llama-2 7B Sauerkraut, Llama-3

Table 1. Performance Metrics of All Tested Models.*			
Model and Metric	Mean	Standard Deviation	Median
Llama-2 Sauerkraut 7B			
Specificity	0.992	0.009	0.9942
False-positive rate	0.008	0.009	0.0058
False-negative rate	0.1582	0.1068	0.1312
Sensitivity	0.8418	0.1068	0.8688
Accuracy	0.9893	0.0092	0.9916
Llama-2 7B			
Specificity	0.9739	0.0174	0.9792
False-positive rate	0.0261	0.0174	0.0208
False-negative rate	0.0494	0.0428	0.0492
Sensitivity	0.9506	0.0428	0.9508
Accuracy	0.9734	0.017	0.9786
Llama-3 8B			
Specificity	0.9814	0.0121	0.983
False-positive rate	0.0186	0.0121	0.017
False-negative rate	0.0058	0.0249	0
Sensitivity	0.9942	0.0249	1
Accuracy	0.9816	0.0119	0.9832
Llama-2 70B			
Specificity	0.9735	0.0192	0.9806
False-positive rate	0.0265	0.0192	0.0194
False-negative rate	0.0115	0.0325	0
Sensitivity	0.9885	0.0325	1
Accuracy	0.9737	0.019	0.9807
Llama-2 "Sauerkraut" 70B			
Specificity	0.9812	0.007	0.9817
False-positive rate	0.0188	0.007	0.0183
False-negative rate	0.0095	0.0286	0
Sensitivity	0.9905	0.0286	1
Accuracy	0.9813	0.0069	0.9818
Llama-3 70B			
Specificity	0.9809	0.0127	0.982
False-positive rate	0.0191	0.0127	0.018
False-negative rate	0.0083	0.0401	0
Sensitivity	0.9917	0.0401	1
Accuracy	0.9811	0.0126	0.9822
Mistral 7B			
Specificity	0.9793	0.0187	0.9825
False-positive rate	0.0207	0.0187	0.0175
False-negative rate	0.0624	0.0571	0.0532
Sensitivity	0.9376	0.0571	0.9468
Accuracy	0.9785	0.0185	0.9818
Phi-3 Mini			
Specificity	0.9674	0.0139	0.9642
False-positive rate	0.0326	0.0139	0.0358
False-negative rate	0.0834	0.1052	0.0538
Sensitivity	0.9166	0.1052	0.9462
Accuracy	0.9664	0.0137	0.9634

*The macro scores of the mean, standard deviation, and median are shown from the anonymization experiments with 250 clinical letters compared with manually annotated data.

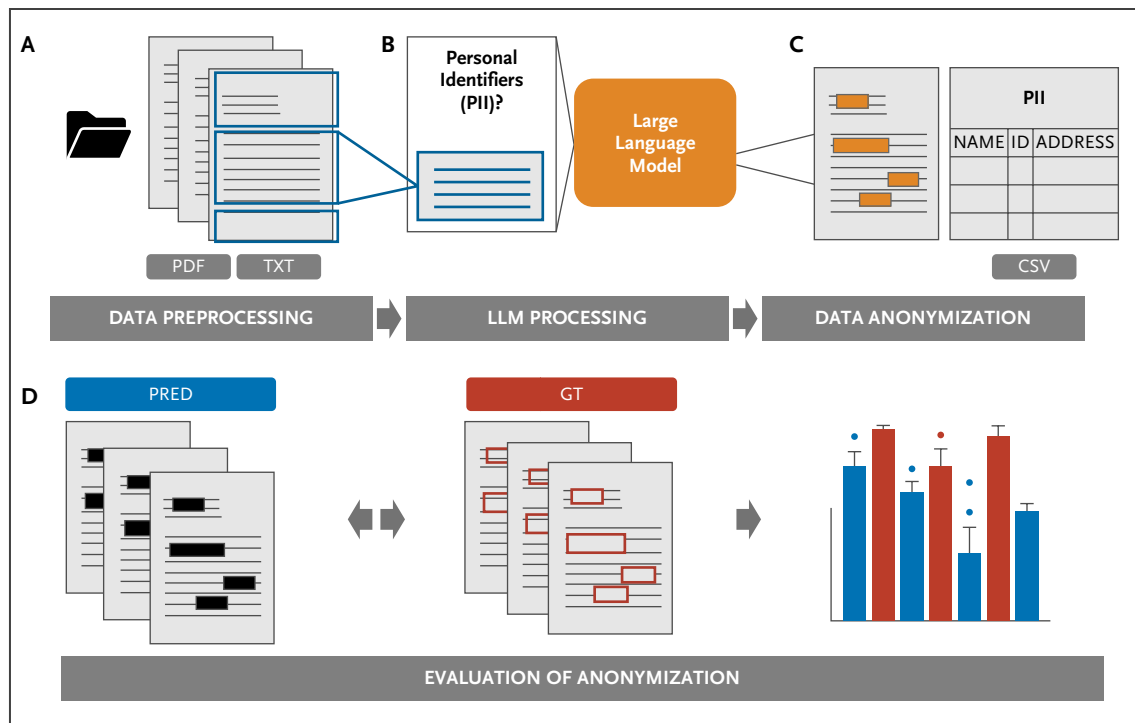


Figure 1. Anonymizer Workflow.

Panel A shows medical documents, such as clinical letters, in PDF and TXT formats, which can be uploaded in the data preprocessing step. The PDF file input format is then converted into raw text, using optical character recognition (OCR) when necessary. The raw text files are then split into chunks of variable character size, which the user can specify. Different chunk sizes may be necessary due to the different context windows of the desired large language models (LLMs). Panel B shows steps after preprocessing; the predefined LLM extracts personal identifiers from the document chunks and transfers them to a CSV file (LLM processing). Panel C shows the personal identifiers are then hidden in a copy of the original input PDF files (data anonymization). Panel D shows the anonymization evaluation; each redacted document (PRED) can be compared with an annotated ground truth dataset (GT), and performance metrics (F1 score, accuracy, sensitivity, precision, specificity, false-positive rate, and false-negative rate) are calculated and displayed for each document as well as for all processed documents. CSV denotes comma-separated value; GT, ground truth; LLM, large language model; PDF, portable document format; PII, personally identifiable information; PRED, prediction; and TXT, text.

70B, Llama-3 8B, Llama-2 Sauerkraut 70B, and Mistral 7B. Llama-2 7B deidentified date of birth accurately in 242 out of 250 letters, whereas Phi-3 Mini did so in 215 out of 250 letters. Llama-3 70B showed overall superior performance, achieving the lowest FNR across all identifier categories (Fig. 4). Although the patient's address and ID typically appeared only once in the dataset, patient name and date of birth appeared four or five times more frequently (Fig. S4).

REPRODUCIBILITY

As reproducibility is a relevant and critical issue for LLM-based experiments, we repeated the experiments with the best model, Llama-3 70B, three times. We found accuracy rates of 98.1, 98.3, and 98.3% for all three model runs, as well as similar sensitivity (99.2, 99.2, and 99.2%). The metrics across all clinical letters were tested for normality

with the Shapiro-Wilk test and found to be not normally distributed. Therefore, we performed the Kruskal-Wallis test to test for statistical significance between all three models' results, which confirmed nonsignificance for the sensitivity comparison ($P=0.92$). The accuracy comparison showed a P value of 0.042, indicating a differing distribution of the three runs' values, which can be attributed to an outlier (Fig. S8).

DETAILED ERROR ANALYSIS

A detailed error analysis for all annotated personal identifiers which were not redacted by the pipeline was performed for the pipeline output of the Llama-3 70B model. The errors were manually reviewed and categorized, with the most frequent error being spelling mistakes of the last name in the free text created by the author of the letter (Fig. S7).

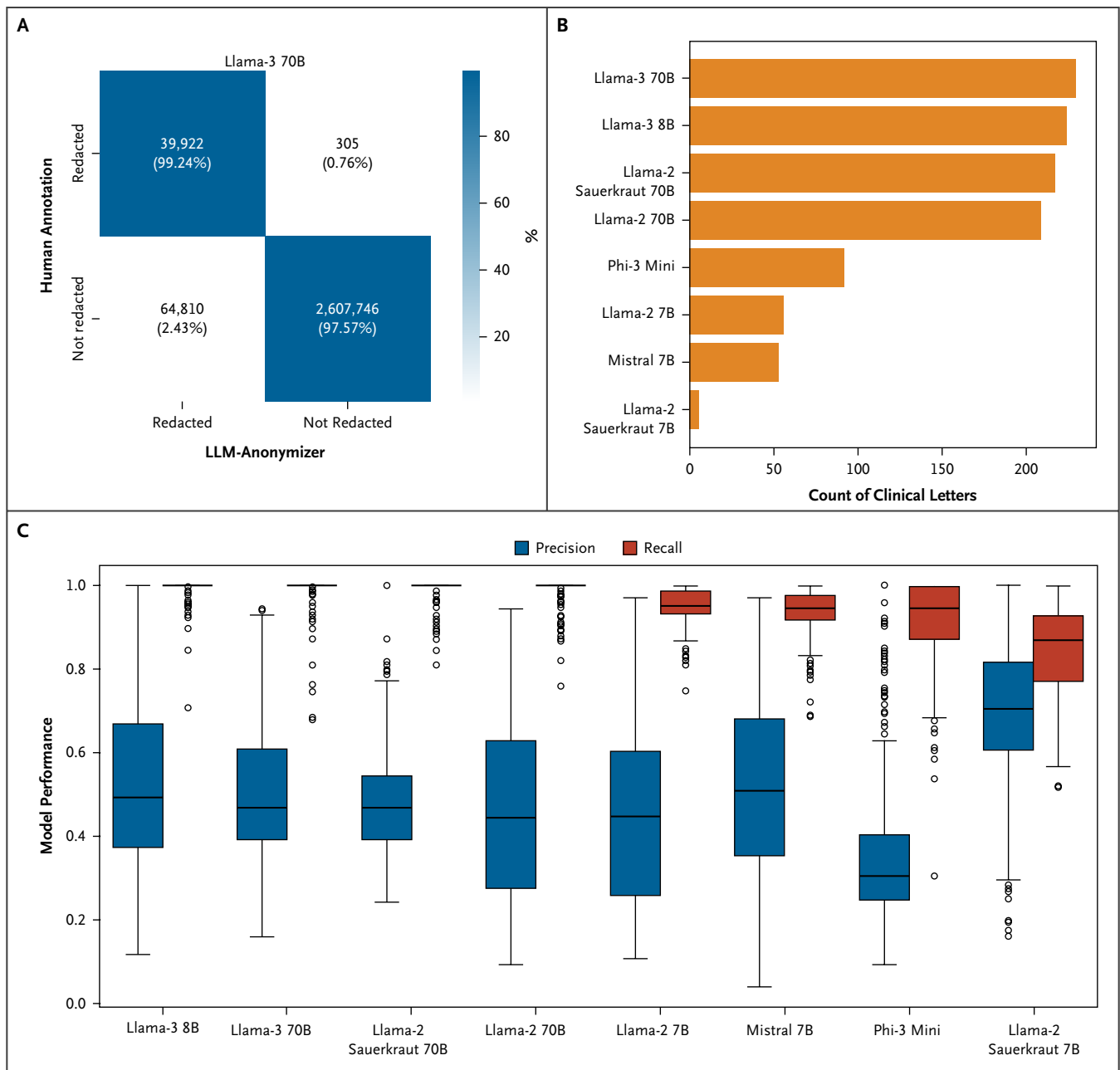


Figure 2. Model Comparison for Anonymization of Medical Documents.

Panel A shows the confusion that the matrix presents the sensitivity and specificity of Llama-3 70B for the deidentification of 250 real-world clinical letters. The LLM-Anonymizer correctly redacted 99.24% of patient-sensitive identifiable information (character wise), with only 0.76% of the information to be redacted being missed and 2.43% being redacted even if not necessary (microscores). B, The Llama-3 models demonstrated the greatest effectiveness in the number of redacted letters, where no potentially sensitive information was missing to be redacted. While Llama-3 70b did not miss any information in 230 letters, the least effective model, Llama-2 Sauerkraut 7B, did not miss any information in less than 10 letters. Panel C shows the box plots illustrating the distribution of the performance metrics, precision and sensitivity, for all tested models. A high sensitivity indicates the percentage of characters that have been correctly redacted, whereas precision demonstrates the percentage of characters containing personal identifying information among all characters redacted. LLM, large language model.

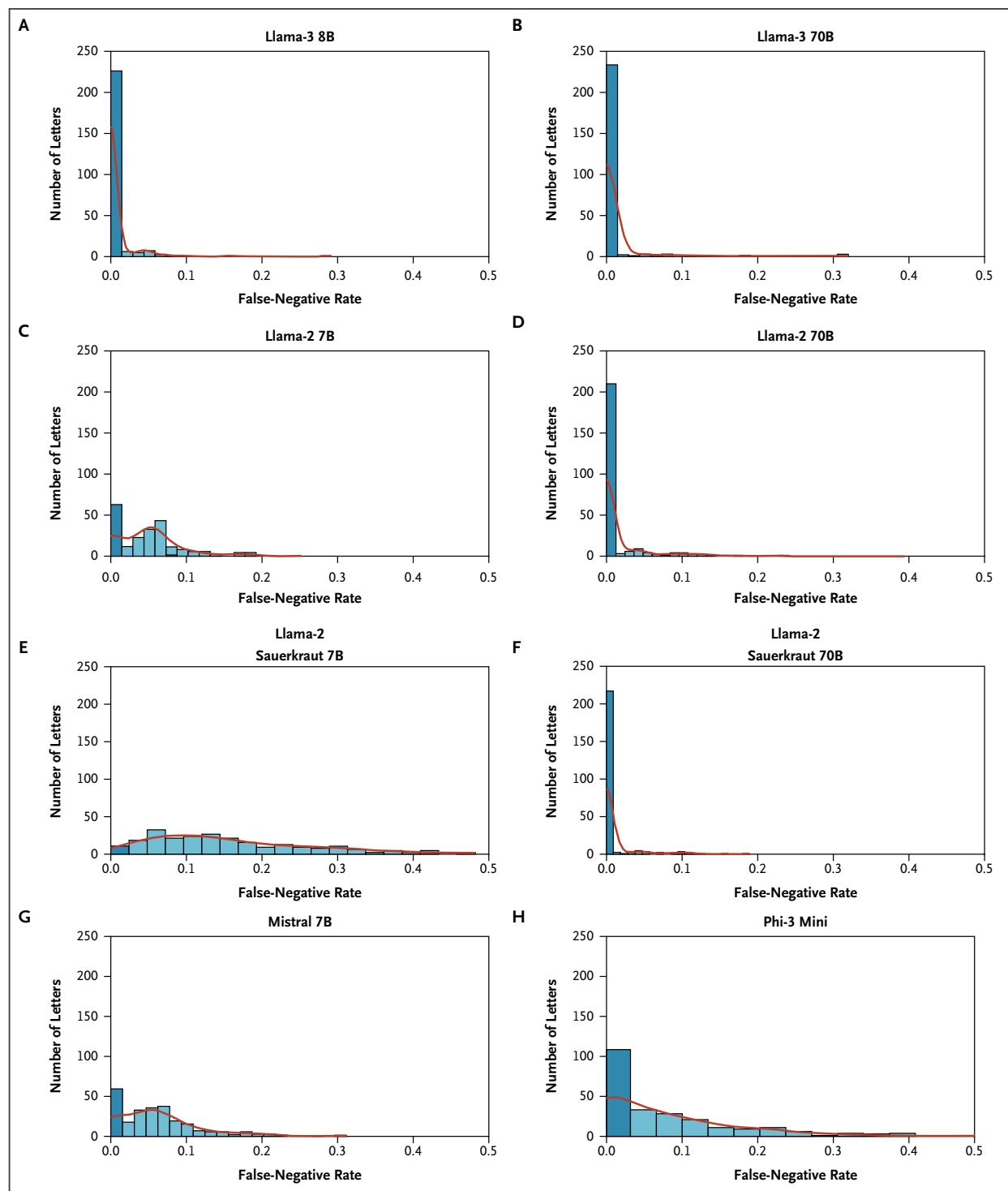


Figure 3. False-Negative Rate per Clinical Letter.

When deidentifying medical documents, the false-negative rate (FNR) per doctor's letter is critical to evaluate the effectiveness of deidentification. The y-axis represents the number of letters redacted by the LLM-Anonymizer, whereas the x-axis represents the FNR. A FNR of zero indicates that all human annotations were redacted in the clinical letters, which are highlighted in dark yellow. The Llama-3 models show the best FNR with a slight improvement from the smaller (8 billion parameters, Panel A) to the larger model (70 billion parameters, Panel B). Panels C and D show the results for the Llama-2 models with parameter sizes of 7 billion and 70 billion, respectively. Panels E and F depict the FNR of the Llama-2-based models, which were fine-tuned on the German-language Sauerkraut model.²² Panels G and H show the FNR for the Mistral 7B and Microsoft's Phi-3 models. FNR, false-negative rate.

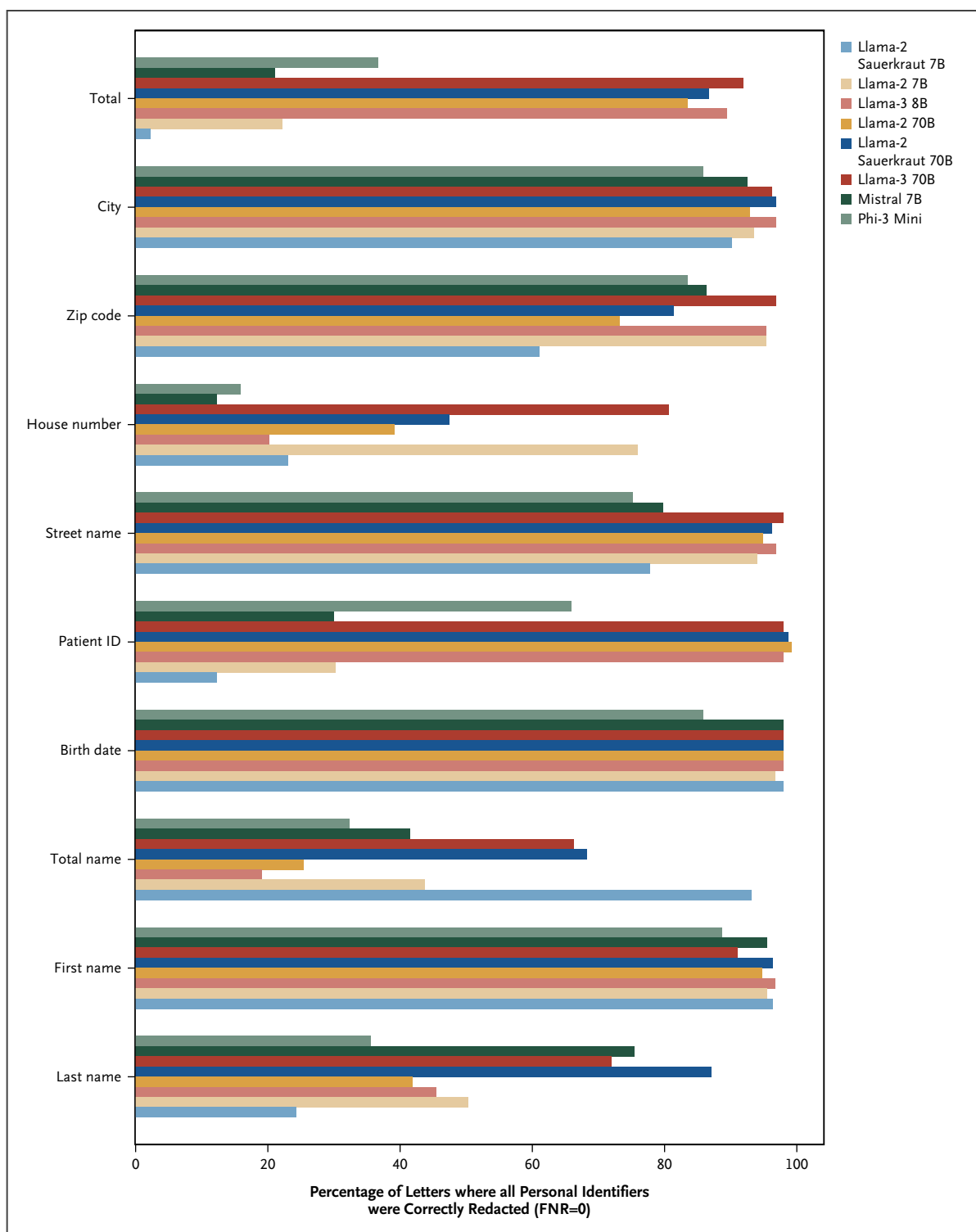


Figure 4. Zero False-Negative Rates for All Models and Labels.

The grouped bar chart shows the percentage of clinical letters with a false-negative rate of zero for each individual label. First name and date of birth were extracted in nearly all letters for almost all models, while house number was more difficult to extract correctly for some models. The y-axis represents all labels for patient identifying information, such as first name, last name, date of birth, city, zip code, patient ID, birth name, and house number. The x-axis represents the percentage of letters for which all label occurrences were redacted (FNR=0). Each color in the grouped bars corresponds to a specific model, including Llama-2 70B, Llama-2 7B, Llama-3 70B, Llama-3 8B, Mistral 7B, Phi-3 Mini, and Llama-2 Sauerkraut in the 70B and 7B variant. FNR, false-negative rate.

INFORMATION EXTRACTION WITH THE LLAMA-3 70B MODEL

The Anonymizer pipeline evaluation was an end-to-end validation, measuring the correctly redacted characters. When considering the information extraction of PII alone, Llama-3 70B achieved a mean accuracy of 58%. These results did not align with the end-to-end validation metrics of the pipeline. The discrepancy arose because the model lacked precise entity definitions (e.g., it extracted only the last name, while our annotation included both given and last names, and missed the academic title). We enhanced the prompt by adding an example to leverage in-context learning. This adjustment improved the performance, resulting in an overall accuracy of 98%. Detailed label-wise metrics are provided in Figure S5.

BENCHMARK ANALYSIS WITH CLINIDEID AND PRESIDIO

To benchmark our anonymizer pipeline, we compared its performance against the open-source anonymization tools CliniDeID (by Clinacuity) and Presidio (by Microsoft) using our annotated dataset. We selected CliniDeID for its dedicated purpose of anonymizing clinical notes and Presidio for its context-aware capabilities, targeting two primary aspects of our tool. Although we aimed to map our PII entities to the corresponding tool-specific internal categories, the comparison was limited due to the use of the premise versions of CliniDeID and Presidio, which rely on predefined PII entities, restricting variable-wise comparisons. Overall, CliniDeID and Presidio anonymized PII in the clinical letters with lower sensitivity (0.84 and 0.82, respectively) on a character-wise basis, with detailed metrics provided in [Table 2](#) and Figure S9. CliniDeID showed poor performance in individual variables but scored well overall because, for example, to its multiple name categories, often misclassifying patient names as provider names. Their high false-positive rates can partly be attributed to broad categorization (e.g., redacting all names, whereas we focused exclusively on patient names).

Conclusion

The removal of PII from medical text data is critical to ensure the privacy of sensitive patient information, but it presents a formidable challenge for data use and sharing. In this article, we benchmarked eight locally deployable LLMs for personal identifier extraction from 250 real-world clinical letters and their subsequent anonymization. In addition, we present the LLM-Anonymizer, a comprehensive,

automated pipeline that streamlines this process for secure deidentification of medical text data on local hardware and with a user-friendly interface, not requiring any programming knowledge.

Existing NLP NER approaches have demonstrated notable performance for anonymizing medical documents (rule based: sensitivity 0.95, precision 0.93).²⁸ Recent advancements in transformer models have showcased similar capabilities in various NLP tasks (precision and sensitivity 0.94).²⁹ LLMs, such as Generative Pre-trained Transformer 4 (GPT-4) have proven to possess advanced anonymization skills.³⁰ LLMs fine-tuned for medical purposes demonstrated a slightly worse performance (precision of 0.91, sensitivity of 0.95) in anonymizing medical documents.²⁸ To our knowledge, we are the first to show the high performance of inference from local LLMs, specifically Llama-2 and -3 models, in extracting PII from medical documents, whereas others achieved only insufficient results. Liu et al. tried zero-shot medical text anonymization with GPT-4 and Llama models, but the Llama models failed to generate any relevant anonymization output for tested medical documents with an accuracy of 0.61. GPT-4 demonstrated a superior accuracy of 0.91 for implicit and 0.99 for explicit shot prompting on their synthetic medical dataset.³¹ Without exploiting the full potential of LLM capabilities, our LLM-Anonymizer achieved comparably high and robust performance in anonymizing medical documents with zero-shot inference for accuracy (Llama-3 70B 99.2) and sensitivity (Llama-3 70B 97.9). In terms of accuracy, the LLM-Anonymizer performs equally to GPT-4, but the strength of our experiments is the local setup and the evaluation of highly varied, real-world medical text data. These results demonstrate that the LLM-Anonymizer is not only competitive with all state-of-the-art (SOTA) NLP techniques in deidentifying medical documents, but also provides a locally deployable solution for real-world medical data.

The LLM-Anonymizer is designed with a flexible definition of anonymizing information, allowing a direct customization of entities within the prompt. This enhances its versatility, especially if specific personal identifiers are meant not to be redacted. It accepts different format types and can also process scanned documents using optical character recognition (OCR). In addition, it supports easy integration of future LLMs, facilitating the selection and exchange of the most suitable models without the need for specific training or fine-tuning, as it operates solely on LLM inference. Further optimization of performance is possible through in-context learning and further prompt engineering,³² which

Table 2. Performance Metrics for Benchmark Analysis.*			
Variable and Metric	LLM-Anonymizer Llama-3 70B	ClinicalDeID	Presidio
Overall			
Sensitivity	0.9917	0.8357	0.8156
Specificity	0.9809	0.8428	0.7038
Accuracy	0.9811	0.8427	0.7056
False-negative rate	0.0083	0.1519	0.1742
False-positive rate	0.0191	0.1498	0.2861
City			
Sensitivity	0.9727	0.0329	0.4190
Specificity	0.9966	0.9955	0.9691
Accuracy	0.9965	0.9944	0.9684
Zip code			
Sensitivity	0.9740	0.0522	0.0080
Specificity	0.9995	0.9998	0.9686
Accuracy	0.9994	0.9992	0.9680
House number			
Sensitivity	0.8474	0.0240	0.0080
Specificity	0.9867	0.9976	0.9686
Accuracy	0.9866	0.9973	0.9684
Street name			
Sensitivity	0.9813	0.0126	0.0342
Specificity	0.9989	0.9976	0.9687
Accuracy	0.9989	0.9960	0.9672
Patient ID			
Sensitivity	0.9800	0.9960	0.00
Specificity	1.000	0.9959	0.9990
Accuracy	1.000	0.9959	1.0000
Birth date			
Sensitivity	0.9973	0.8465	0.9973
Specificity	0.9996	0.9761	0.9742
Accuracy	0.9996	0.9756	0.9743
Total name			
Sensitivity	0.8499	0.1094	0.9304
Specificity	0.9979	0.9922	0.7595
Accuracy	0.9969	0.9969	0.7605
First name			
Sensitivity	0.9368	0.1138	0.9311
Specificity	0.9986	0.9919	0.7573
Accuracy	0.9984	0.9895	0.7577
Last name			
Sensitivity	0.9121	0.1064	0.9094
Specificity	0.9966	0.9921	0.7583
Accuracy	0.9961	0.9883	0.7589

*The macro scores are shown for the anonymization task using the LLM-Anonymizer with Llama-3 70B, CliniDeID, and Presidio, based on 250 clinical letters, compared with manually annotated data, and broken down into overall scores and scores for individual variables.

is explored by ongoing projects. Our LLM-Anonymizer enables the comparison of annotation quality against manually annotated datasets. It provides comprehensive metrics

for the entire document set and individual documents. This allows research teams from different locations to upload their own ground truth to evaluate the LLM-Anonymizer

on their specific dataset. The tool-assisted review of each document enables the rapid assessment of redaction quality and the iteration of deidentification entities for the purpose of process improvement, offering substantial flexibility and variability in addressing the critical task of document anonymization. We focused on sensitivity and counts of letters with zero false negatives for the feasibility assessment, because removing all personal identifiers completely is more critical than mistakenly removing nonpersonal information, as long as the original meaning of the text remains accessible. However, this approach may have resulted in a trade-off between precision and sensitivity. Precision might have suffered because our redacted entity definitions were too fine-grained for some labels. For instance, the indiscriminate redaction of numerical entities, such as house numbers, when extracted independently rather than in context with other address details, could also mistakenly remove important laboratory values and other numerical data. In the benchmark analysis, our anonymizer tool outperformed the anonymization tools CliniDeID and Presidio in anonymizing PII within our annotated dataset. CliniDeID has the advantage over our tool of being HIPAA-compliant, covering a broad range of categories, which limits the comparison; however, as our PII entities are included in the official list of HIPAA identifiers, they can, therefore, be directly compared. Regarding individual entities, the LLM-Anonymizer demonstrated higher sensitivity in correctly detecting them. A key strength of our tool is its flexibility, particularly in handling context-dependent entities, enabling it to quickly adapt to new rules — an essential feature for processing medical text data. Unlike rule-based tools, this approach effectively manages diverse entity types, such as German names and street names or patient IDs, without predefined structures, providing a more robust solution.

The degree of anonymization through redaction of PII is in itself the subject of research.³³ Even if all common personal identifying attributes are omitted, the remaining information might still be sufficient to re-identify the individual.³⁴ In addition, the documentation of medical texts exhibits significant variability due to inconsistent record-keeping practices, variations in patient names and misspellings, which present challenges for traditional NER techniques in accurately identifying personal information.³⁵ LLMs offer a special opportunity here; they are capable of grasping concepts rather than just keywords. Misspellings and inconsistent record-keeping practices do not pose a substantial concern. They may even be able to identify all possible identifying, more abstract entities in a multilevel approach, before these entities can be extracted for anonymization. This needs to be clarified in further research.

With regard to future improvements of our pipeline, the next steps incorporate extending the evaluation toward all criteria as defined by HIPAA for full anonymization, including the personal information of treating physicians. Our study is designed to comply with European GDPR, which serves as the EU counterpart to HIPAA. The GDPR views anonymization not as a binary process but as a continuum, where the goal is to reduce the risk of reidentification below a certain threshold. This means that while personal information should be removed, the document must remain usable for its intended purpose. Although our dataset is diverse and encompasses various medical document formats and content, its size remains constrained. The shape of medical documents, including the revealed PII, varies across hospitals and their informatics structure. Consequently, the generalizability of our findings needs to be evaluated on larger and more generalized, annotated datasets sourced from diverse documents of multiple health care facilities.

Despite numerous opportunities for improvement, we were able to show that LLMs are promising in identifying patient-sensitive data and present a valuable, easy-to-use, open-source tool that simplifies the deidentification of medical text data, facilitating its cross-site use.

Disclosures

Author disclosures and other supplementary materials are available at ai.nejm.org

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The data used in this study contain patient-sensitive information and will, therefore, not be published. We provide fictitious examples of clinical letters in the Supplementary Appendix.

All code for the LLM-Anonymizer is freely available under an open-source license at <https://github.com/KatherLab/LLMAnonymizer-Publication>

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References

1. Häyrinen K, Saranto K, Nykänen P. Definition, structure, content, use and impacts of electronic health records: a review of the research literature. *Int J Med Inform* 2008;77:291-304. DOI: [10.1016/j.ijmedinf.2007.09.001](https://doi.org/10.1016/j.ijmedinf.2007.09.001).
2. Naumann L, Esdar M, Ammenwerth E, Baumberger D, Hübner U. Same goals, yet different outcomes: analysing the current state of eHealth adoption and policies in Austria, Germany, and Switzerland using a mixed methods approach. *Stud Health Technol Inform* 2019;264:1012-1016. DOI: [10.3233/SHT1190377](https://doi.org/10.3233/SHT1190377).
3. Tsai CH, Eghdam A, Davoody N, Wright G, Flowerday S, Koch S. Effects of electronic health record implementation and barriers to adoption and use: a scoping review and qualitative analysis of the content. *Life* 2020;10:327. DOI: [10.3390/life10120327](https://doi.org/10.3390/life10120327).
4. Kierkegaard P. Electronic health record: wiring Europe's health-care. *Comput Law Secur Rev* 2011;27:503-515. DOI: [10.1016/j.clsr.2011.07.013](https://doi.org/10.1016/j.clsr.2011.07.013).
5. Uslu A, Stausberg J. Value of the electronic medical record for hospital care: update from the literature. *J Med Internet Res* 2021;23:e26323. DOI: [10.2196/26323](https://doi.org/10.2196/26323).
6. Perez-Lopez R, Ghaffari Laleh N, Mahmood F, Kather JN. A guide to artificial intelligence for cancer researchers. *Nat Rev Cancer* 2024;24:427-441. DOI: [10.1038/s41568-024-00694-7](https://doi.org/10.1038/s41568-024-00694-7).
7. Singhal K, Azizi S, Tu T, et al. Large language models encode clinical knowledge. *Nature* 2023;620:172-180. DOI: [10.1038/s41586-023-06291-2](https://doi.org/10.1038/s41586-023-06291-2).
8. Wiest IC, Ferber D, Zhu J, et al. From text to tables: a local privacy preserving large language model for structured information retrieval from medical documents. December 8, 2023 (<https://www.medrxiv.org/content/10.1101/2023.12.07.23299648.abstract>). Preprint.
9. Adams LC, Truhn D, Busch F, et al. Leveraging GPT-4 for post hoc transformation of free-text radiology reports into structured reporting: a multilingual feasibility study. *Radiology* 2023;307:e230725. DOI: [10.1148/radiol.230725](https://doi.org/10.1148/radiol.230725).
10. Truhn D, Loeffler CM, Müller-Franzes G, et al. Extracting structured information from unstructured histopathology reports using generative pre-trained transformer 4 (GPT-4). *J Pathol* 2024;262:310-319. DOI: [10.1002/path.6232](https://doi.org/10.1002/path.6232).
11. Ferber D, Wiest IC, Wölflin G, et al. GPT-4 for information retrieval and comparison of medical oncology guidelines. *NEJM AI* 1. DOI: [10.1056/AIcs2300235](https://doi.org/10.1056/AIcs2300235).
12. Kather JN, Ferber D, Wiest IC, Gilbert S, Truhn D. Large language models could make natural language again the universal interface of healthcare. *Nat Med* 2024;30:2708-2710. DOI: [10.1038/s41591-024-03199-w](https://doi.org/10.1038/s41591-024-03199-w).
13. Moor M, Banerjee O, Abad ZSH, et al. Foundation models for generalist medical artificial intelligence. *Nature* 2023;616:259-265. DOI: [10.1038/s41586-023-05881-4](https://doi.org/10.1038/s41586-023-05881-4).
14. Rösler W, Altenbuchinger M, Baeßler B, et al. An overview and a roadmap for artificial intelligence in hematology and oncology. *J Cancer Res Clin Oncol* 2023;149:7997-8006. DOI: [10.1007/s00432-023-04667-5](https://doi.org/10.1007/s00432-023-04667-5).
15. Meystre SM, Friedlin FJ, South BR, Shen S, Samore MH. Automatic de-identification of textual documents in the electronic health record: a review of recent research. *BMC Med Res Methodol* 2010;10:70. DOI: [10.1186/1471-2288-10-70](https://doi.org/10.1186/1471-2288-10-70).
16. Zuo Z, Watson M, Budgen D, Hall R, Kennelly C, Al Moubayed N. Data anonymization for pervasive health care: systematic literature mapping study. *JMIR Med Inform* 2021;9:e29871. DOI: [10.2196/29871](https://doi.org/10.2196/29871).
17. Neamatullah I, Douglass MM, Lehman L-WH, et al. Automated de-identification of free-text medical records. *BMC Med Inform Decis Mak* 2008;8:32. DOI: [10.1186/1472-6947-8-32](https://doi.org/10.1186/1472-6947-8-32).
18. Kushida CA, Nichols DA, Jadrnicek R, Miller R, Walsh JK, Griffin K. Strategies for de-identification and anonymization of electronic health record data for use in multicenter research studies. *Med Care* 2012;50:S82-S101. DOI: [10.1097/MLR.0b013e3182585355](https://doi.org/10.1097/MLR.0b013e3182585355).
19. Gerganov G. llama.cpp. GitHub. 2023 (<https://github.com/ggerganov/llama.cpp>).
20. TheBloke. TheBloke/Llama-2-7B-GGUF. Huggingface (<https://huggingface.co/TheBloke/Llama-2-7B-GGUF>).
21. TheBloke. TheBloke/Llama-2-70B-GGUF. Huggingface (<https://huggingface.co/TheBloke/Llama-2-70B-GGUF>).
22. TheBloke. TheBloke/SauerkrautLM-70B-v1-GGUF. Huggingface (<https://huggingface.co/TheBloke/SauerkrautLM-70B-v1-GGUF>).
23. QuantFactory. QuantFactory/Meta-Llama-3-8B-Instruct-GGUF. Huggingface (<https://huggingface.co/QuantFactory/Meta-Llama-3-8B-Instruct-GGUF/tree/main>).
24. QuantFactory. QuantFactory/Meta-Llama-3-70B-Instruct-GGUF. Huggingface. 2024 (<https://huggingface.co/QuantFactory/Meta-Llama-3-70B-Instruct-GGUF>).
25. TheBloke. TheBloke/Mistral-7B-v0.1-GGUF. Huggingface (<https://huggingface.co/TheBloke/Mistral-7B-v0.1-GGUF>).
26. Microsoft. Microsoft/Phi-3-mini-4k-instruct-gguf. Huggingface (<https://huggingface.co/microsoft/Phi-3-mini-4k-instruct-gguf>).
27. Klie J-C, Bugert M, Boullosa B, Eckart de Castilho R, Gurevych I. The INCEpTION platform: machine-assisted and knowledge-oriented interactive annotation. In: Zhao D, ed. *Proceedings of the 27th International Conference on Computational Linguistics: System Demonstrations*, Association for Computational Linguistics, Santa Fe, NM, August 20-26, 2018:5-9.
28. Azzouzi ME, Coatrieux G, Bellafqira R, et al. Automatic de-identification of French electronic health records: a cost-effective approach exploiting distant supervision and deep learning models. *BMC Med Inform Decis Mak* 2024;24:54. DOI: [10.1186/s12911-024-02422-5](https://doi.org/10.1186/s12911-024-02422-5).

29. Asimopoulos D, Siniosoglou I, Argyriou V, et al. Benchmarking advanced text anonymisation methods: a comparative study on novel and traditional approaches. April 22, 2024 (<http://arxiv.org/abs/2404.14465>). Preprint.
30. Staab R, Vero M, Balunović M, Vechev M. Large language models are advanced anonymizers. February 21, 2024 (<http://arxiv.org/abs/2402.13846>). Preprint.
31. Liu Z, Huang Y, Yu X, et al. DeID-GPT: zero-shot medical text de-identification by GPT-4. March 20, 2023 (<http://arxiv.org/abs/2303.11032>). Preprint.
32. Nori H, Lee YT, Zhang S, et al. Can generalist foundation models outcompete special-purpose tuning? Case Study in Medicine. November 28, 2023 (<http://arxiv.org/abs/2311.16452>). Preprint.
33. Li X-B, Qin J. Anonymizing and sharing medical text records. *Inf Syst Res* 2017;28:332-352. DOI: [10.1287/isre.2016.0676](https://doi.org/10.1287/isre.2016.0676).
34. Patsakis C, Lykousas N. Man vs the machine in the struggle for effective text anonymisation in the age of large language models. *Sci Rep* 2023;13:16026. DOI: [10.1038/s41598-023-42977-3](https://doi.org/10.1038/s41598-023-42977-3).
35. Kundeti SR, Vijayananda J, Mujjiga S, Kalyan M. Clinical named entity recognition: challenges and opportunities. *Proceedings of the IEEE International Conference on Big Data (Big Data)*, IEEE, Washington, DC, December 5-8, 2016:1937-1945.